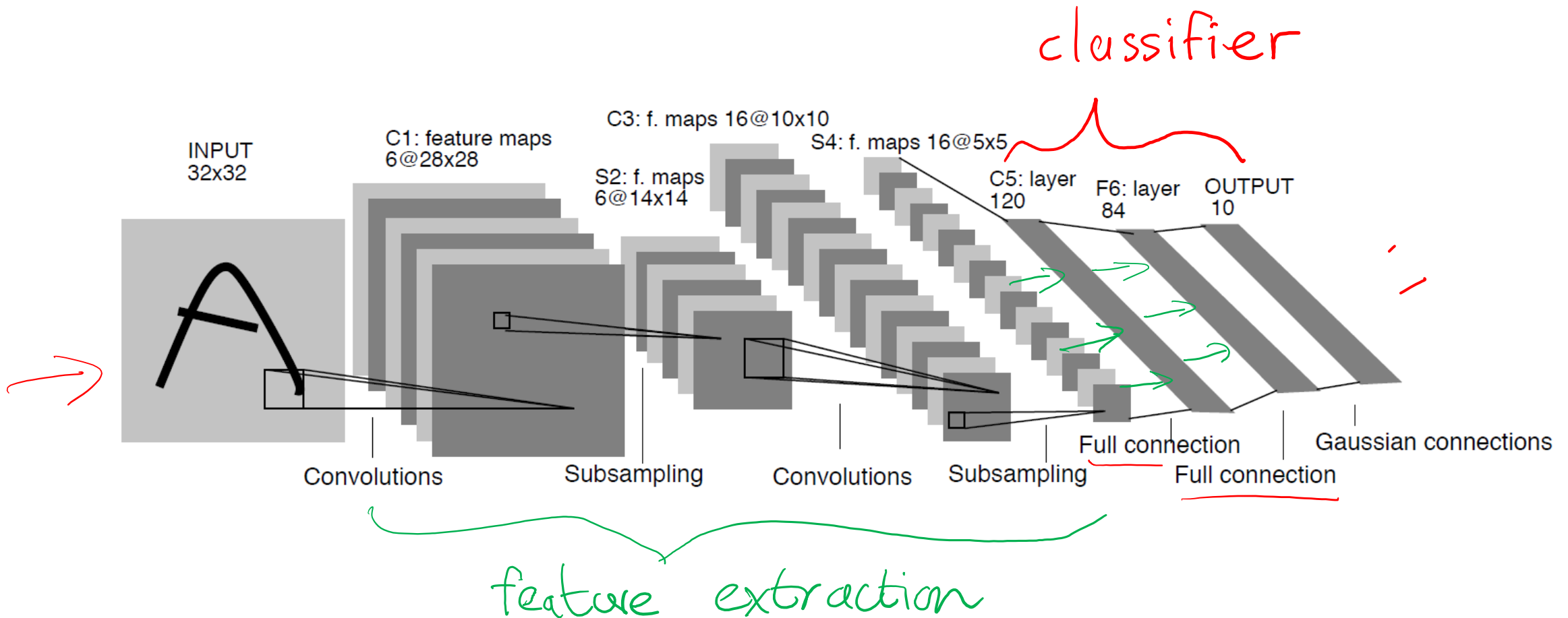


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Intro to Convolutional Neural Networks

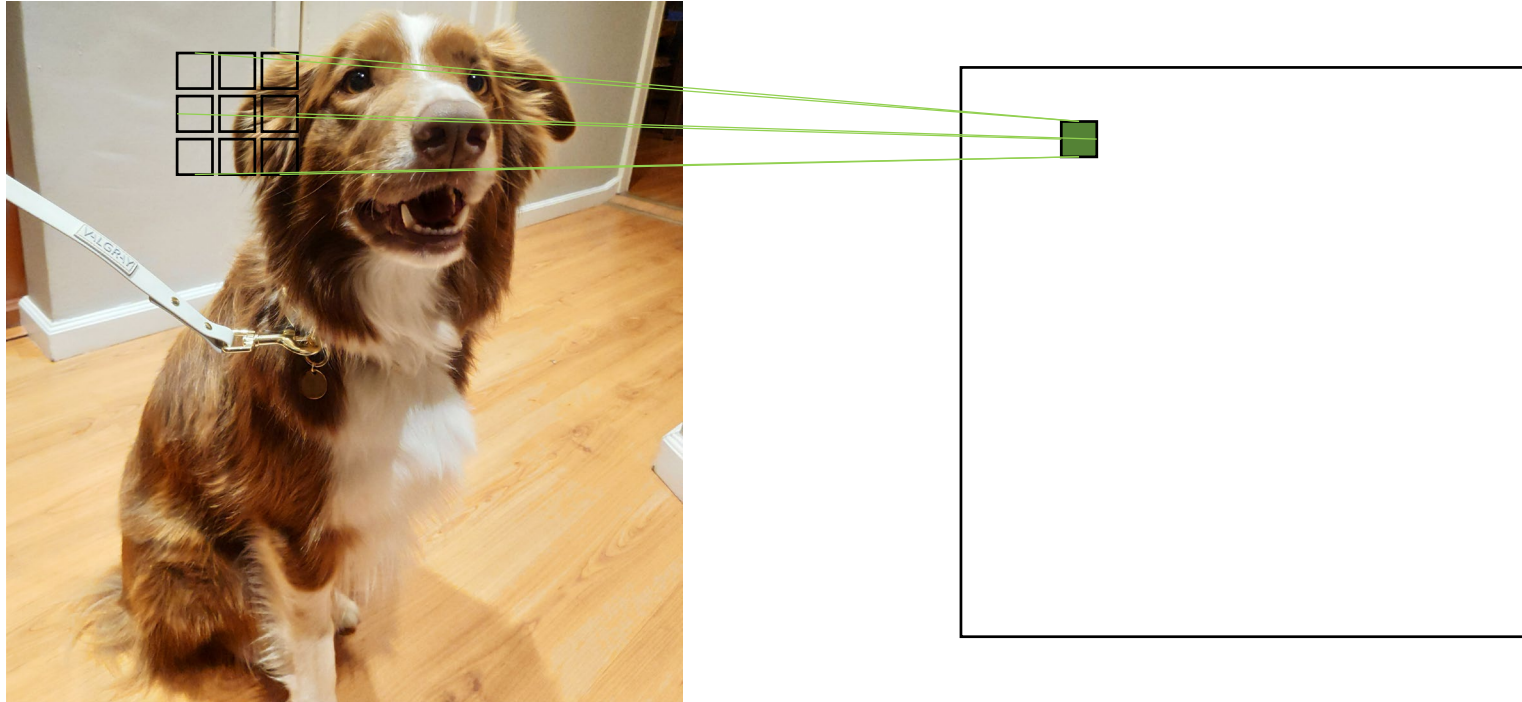


Convolutional Neural Networks

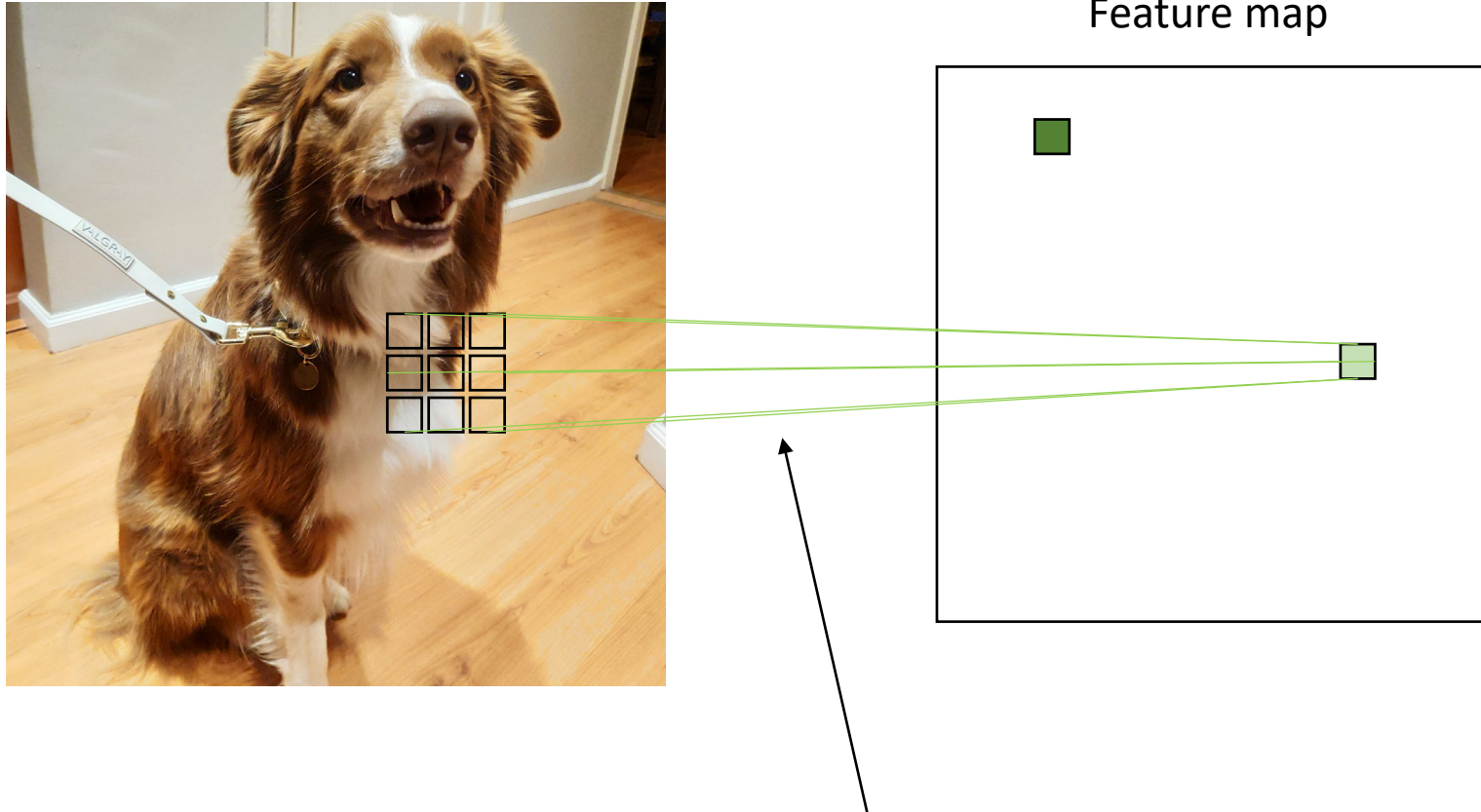


Yann LeCun, Léon Bottou, Yoshua Bengio and Patrick Haffner: Gradient Based Learning Applied to Document Recognition, Proceedings of IEEE, 86(11):2278-2324, 1998.

“Convolution”



“Convolution”



This collection of weights is called a kernel/filter/feature detector

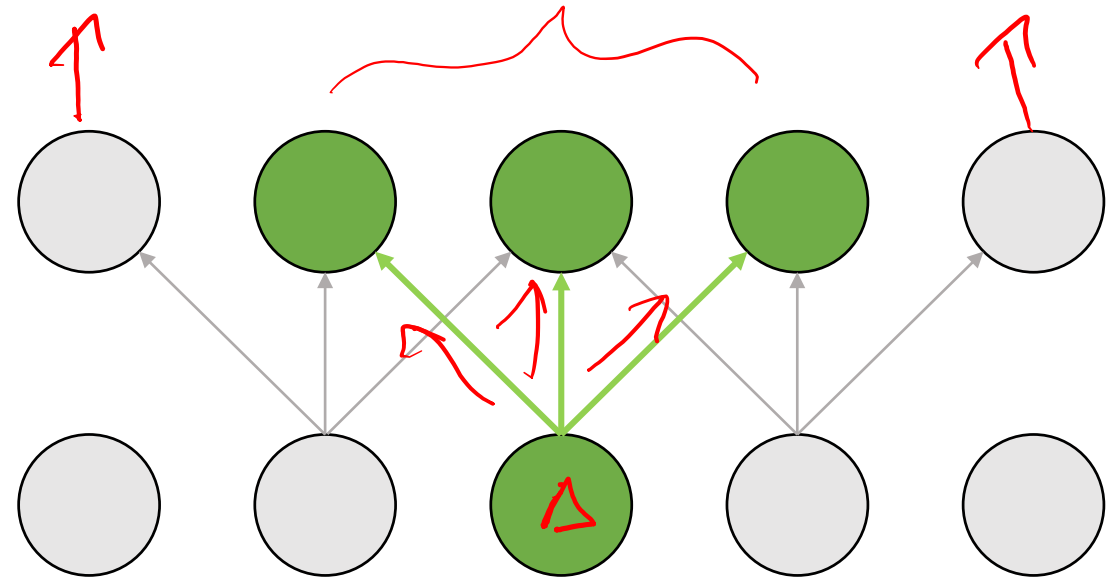
Why?

- **Sparse interactions**
 - Focus on localized regions
- **Parameter Sharing**
 - Learn a small set of reusable weights
- **Equivariant representations**
 - If the input changes the output changes in the same way

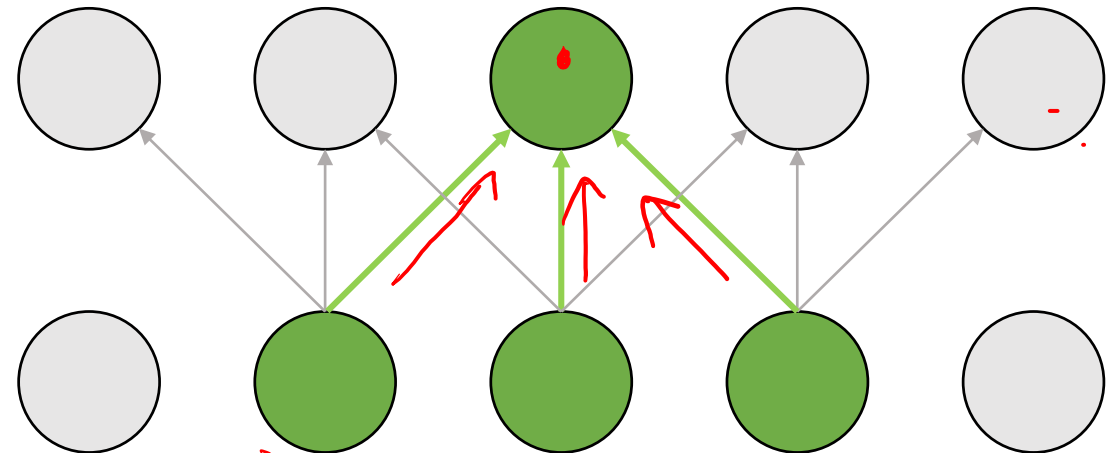


Sparse Interactions

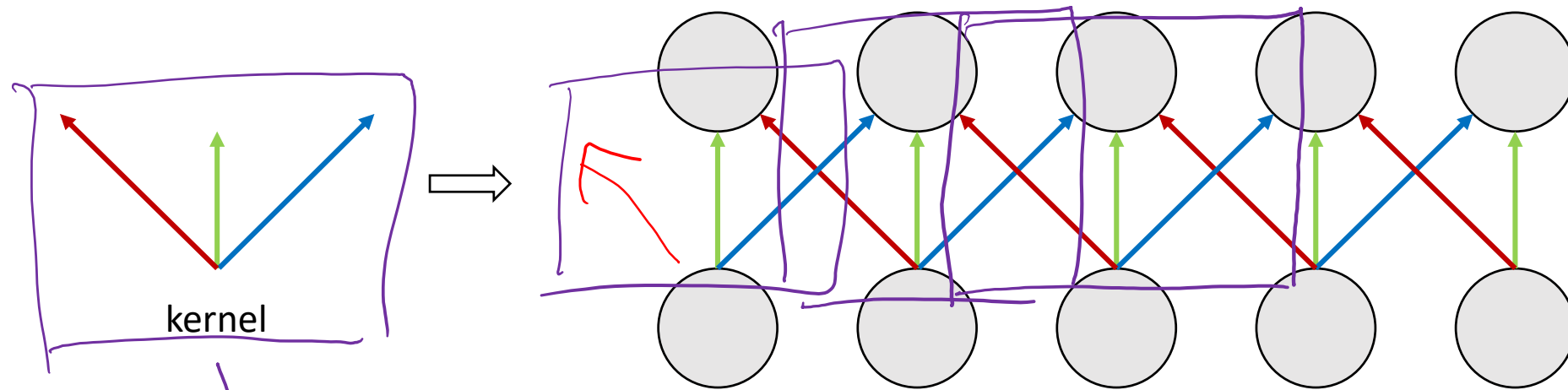
From the bottom up:
One input unit only has an effect on a subset of the units in the next layer



From the top down:
A single unit in the “upper” layer is only affected by a subset of units in the previous layer (receptive field).



Parameter Sharing

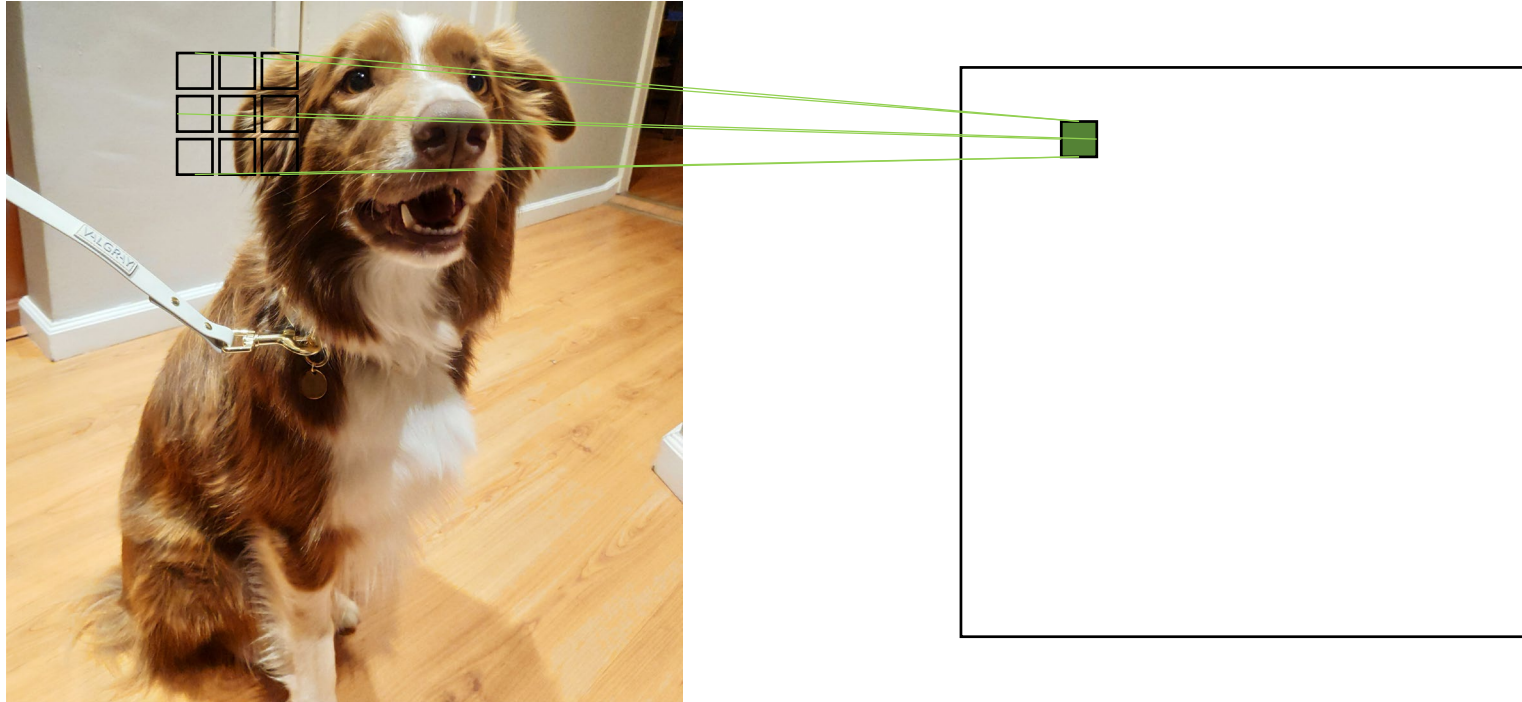


Shared parameters = reduced storage, “less” complexity

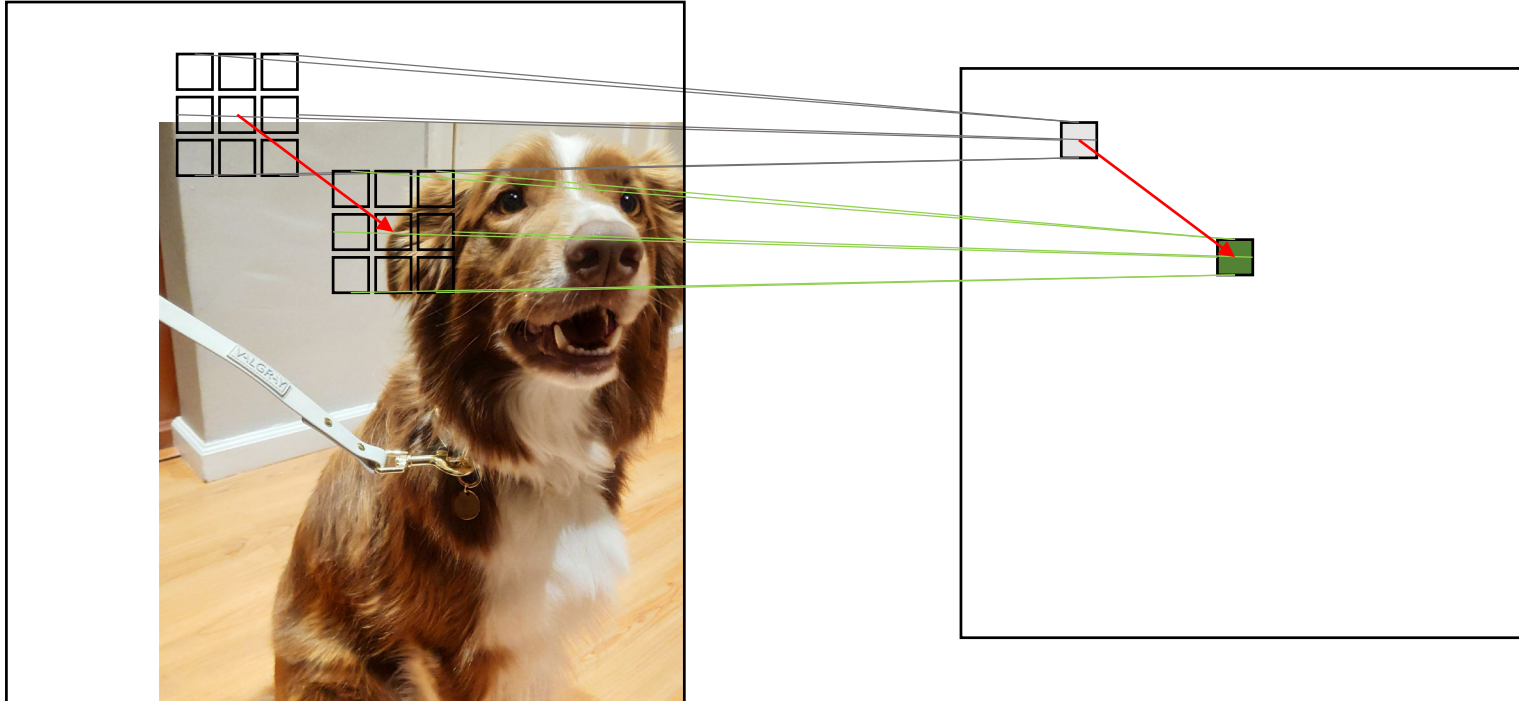
3 values to store compared to $5 \times 5 = 25$ values if this was fully connected



Equivariance to translation



Equivariance to translation



Next up:

Parameter sharing details

